

NEW RESEARCH

DIMENSIONAL APPROACHES

Latent Profile Analysis of Parent-Reported Mood Symptoms in Youth: A Person-Centered Approach to Classification

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Objective: To evaluate whether latent profile analysis (LPA) of dimensional mood symptoms can identify clinically meaningful subgroups of youth, using a measure that captures both positive (manic) and negative (depressive) valence, and to assess convergence with psychiatric diagnoses and interview-based ratings of mood symptom severity.

Method: We conducted a secondary analysis of data from 1,369 treatment-seeking youths (5-18 years of age) recruited from an urban community mental health center and an academic medical center. Consensus *DSM* diagnoses were derived from Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS) interviews, family history, and treatment records. LPA was applied to 20 parent-rated symptom facets from the Parent General Behavior Inventory (PGBI), modeling 2 continuous mood dimensions. Profiles were compared against mood diagnoses and symptom severity based on the Children's Depression Rating Scale—Revised (CDRS-R) and the Young Mania Rating Scale (YMRS), both scored via direct parent-youth interviews by trained clinicians (with PGBI scores masked).

Results: A 5-profile solution was supported by fit indices (Bayesian information criterion, entropy) and interpretability (minimum cluster size: $n = 123$). Profile membership was strongly associated with consensus diagnoses ($\chi^2[16] = 579.39, p < .000005$) and current mood state ratings ($\chi^2[20] = 583.99, p < .000005$). Profiles varied significantly in rates of depressive, manic, and mixed states, and in average CDRS-R and YMRS scores, as well as global functioning and family history of bipolar disorder.

Conclusion: Latent profiles based on dimensional ratings of positive and negative mood symptoms aligned well with structured interview-derived diagnoses and symptom states. These profiles may offer a data-driven method to enhance mood disorder classification and to support early identification efforts in youth psychiatry.

Key words: bipolar disorder; depression; mood episode; latent profile analysis; person-centered analysis

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Psychoiatric diagnosis is increasingly being reimagined in terms of major transdiagnostic dimensions of symptomatology and functioning. The *Diagnostic and Statistical Manual of Mental Disorders (DSM)*,¹ the *International Classification of Diseases (ICD)*,² the Hierarchical Taxonomy of Psychopathology (HiToP)³ and Research Domain Criteria (RDoC)^{4,5} all have embraced dimensional models of psychopathology. Multiple lines of evidence support dimensional models, including statistical analyses such as taxometrics and latent profile models, as well as genetic, neurocognitive, imaging, and treatment response data.

However, mood disorders require attention to multiple dimensions, making them intrinsically more challenging to conceptualize within a dimensional framework. Depression

involves both high levels of negative valence and low levels of positive valence,⁶⁻⁸ whereas classic manic states involve the opposite (low negative valence and high positive valence), and mixed states involve simultaneous elevations of both.⁹ There are likely other major dimensions that factor into the presentation of mood disorders, including disturbances of cognitive functioning and, at least in a subset of cases, psychotic features such as hallucinations and delusions. Viewed through an RDoC lens, disruption of regulatory processes such as circadian rhythms also appears to be promising to consider.

An additional complication is that, by definition, the presentation of mood disorders is episodic, and the episodes can vary in phenomenology or polarity over the longitudinal course of the condition. The *DSM*¹ and *ICD*²

classify mood disorders using lifetime history of mood episodes, with more recent versions emphasizing disturbance of energy and activity as well as mood. Within these nosologies, sleep and cognitive disturbance are embedded in the context of episodic changes in mood and energy. The diagnostic criteria, as operationalized in structured interviews or rating scales, emphasize the episodic presentation and embeddedness in a mood context. Although there have been dimensional measures of depressive and manic symptoms available for decades, including interviewer-based ratings such as the Young Mania Rating Scale (YMRS)¹⁰ and the Child Depression Rating Scale—Revised (CDRS-R)¹¹ that have achieved widespread use, including use as the primary outcome measure in clinical trials and studies for filing with regulatory agencies, as well as self- and cross-informant checklists and rating scales, the operational definitions of mood states on these tools originated as rules of thumb that have become reified without formal empirical investigation. Stated another way, the thresholds separating hypomania, mania, depression, and mixed states have been guided more by convention¹² than by data.

Latent profile analysis (LPA) offers an empirical framework to use dimensional mood scores to identify clusters of youths with similar current mood states.¹³ Modeling both positive- and negative-valenced scores should identify groups with distinct profiles. This paper applies LPA to a well-validated measure with broad coverage of both mood dimensions, comparing classification consistency against psychiatric diagnoses and mood states. We focus on the Parent version of the General Behavior Inventory (PGBI) as the input variable for the LPA, for the following reasons: (1) it has performed in the top tier in meta-analyses of classification accuracy for mood disorders^{14,15}; (2) it is low cost (free to use, with non-proprietary scoring) and has been translated into a variety of languages, thereby enhancing its potential clinical utility; (3) it not only includes all of the *DSM/ICD* symptoms of mania and major depression, but it also assesses a broad set of associated features¹⁶; and (4) prior work has established 20 unifactorial mini-scales, or “item parcels,” that provide technical advantages over directly analyzing 73 items, while offering much more granularity than working with 2 or 3 summary scales.^{15,17} The more fine-grained approach using 20 parcels allows exploration of whether specific symptom facets are more helpful in differentiating qualitatively different profiles of mood symptoms. Looking at which facets are most strongly associated with profile membership could inform debates in the literature, such as whether atypical features are more characteristic of bipolar

depression, as well as whether some symptoms are more implicated in mixed mood states.^{18,19}

We hypothesized that the LPA would identify at least 1 profile without significant mood disorders or clinically elevated mood symptoms, and also expected to have a unipolar depressive profile and at least 1 smaller bipolar profile. We also expected to find at least 1 profile with moderate mood symptoms, which are less easy to label or to fit into the major mood diagnoses in the *DSM*, as attested by the lack of labels for minor (as distinct from major) depression, as well as the extensive discussions about definitions of mixed mood states,²⁰ or various additional prototypical mood presentations.²¹ Angst²² proposed a model of mood disorder and terminology using brief labels to characterize mood presentations, as follows: *M* for mania, *D* for major depression, *m* for hypomania, *d* for minor depression (not a *DSM* category), *md* for cyclothymic disorder and temperament (again, in contrast to *DSM*), and *dsx*, *msx*, and *mdsx* for subthreshold mood symptoms (to distinguish from “supernormals” who were devoid of mood symptoms entirely). In the Angst framework, we would expect to see *mdsx* and *md* presentations to be well represented in a treatment-seeking sample.

METHOD

Participants

The present study conducted a secondary analysis of 2 datasets: the Stanley dataset²³ and the ABACAB dataset (Assessing Bipolar: A Community Academic Blend; NIH R01MH066647).²⁴ These 2 datasets contained comprehensive diagnostic and clinical data for youth 5 to 18 years of age who were seeking outpatient mental health services. The Stanley sample was recruited from an academic medical center, whereas the ABACAB sample comprised 80% of participants recruited from a community mental health center and 20% from the same academic center. Together, these datasets represented a diverse range of clinical presentations and demographics of children and adolescents with mental health concerns.¹³

Diagnoses

Both the Stanley and ABACAB studies used semi-structured diagnostic interviews (Kiddie Schedule for Affective Disorders and Schizophrenia [KSADS])²⁵ and sequential interviews with the youth and their primary caregiver. At a consensus meeting led by a licensed clinical psychologist or board-certified child psychiatrist, KSADS findings were reviewed and integrated with prior treatment history and family psychiatric history, achieving a “LEAD”

(Longitudinal Expert using All Data) diagnosis²⁶ following *DSM-IV* criteria.²⁷ Interrater reliability was kappa > 0.80 for diagnoses of depression, bipolar disorders, attention-deficit/hyperactivity disorder (ADHD), substance use disorders, oppositional defiant disorder (ODD)/conduct disorder (CD), and posttraumatic stress disorder (PTSD).^{23,24}

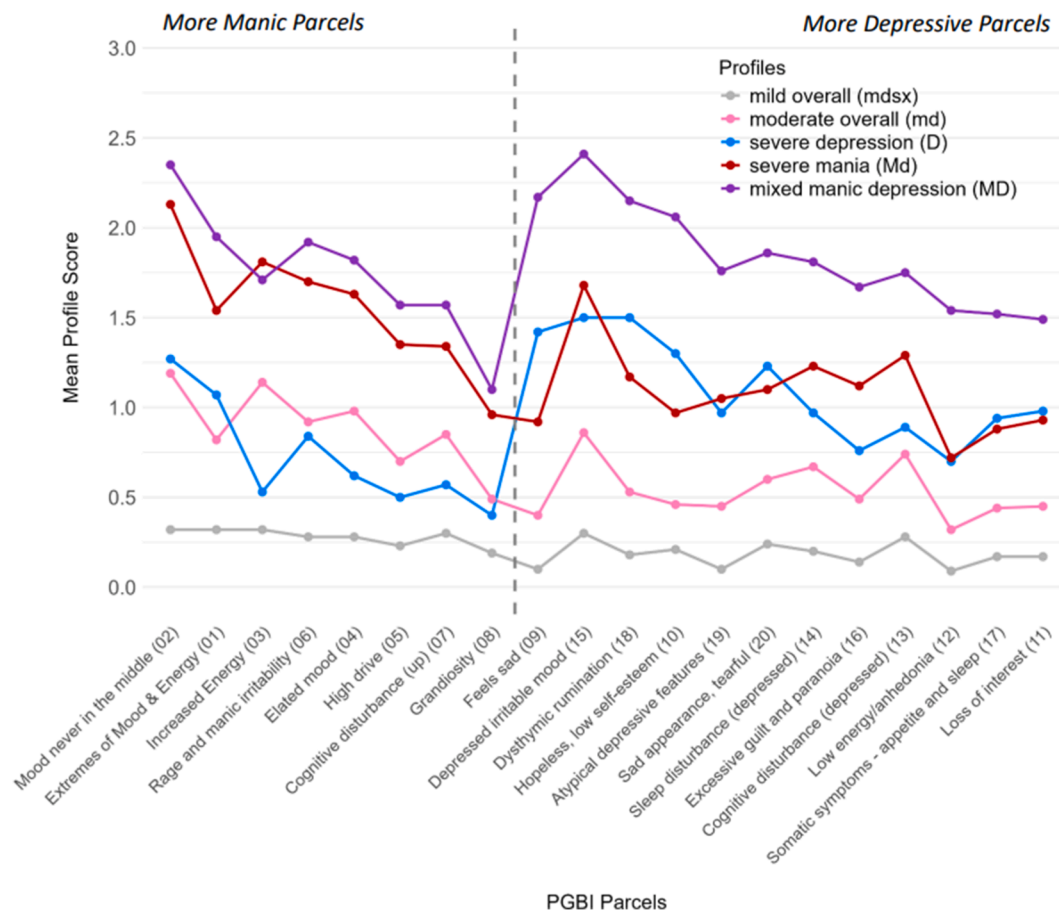
Measures

Indicators for Latent Profiles. LPA used 73 items from the Parent General Behavior Inventory (PGBI).¹⁵ The PGBI is a parent report measure assessing depressive, hypomanic, and mixed symptoms, and it has shown strong discriminative validity for mood disorders in multiple meta-analyses, as well as prognostic validity and treatment sensitivity.^{15,28} The LPA used 20 unifactorial parcels (scaled as item averages) as the indicators of 3 or 4 items, following prior work.^{13,17}

Criterion Validity Indicators. We used demographic and clinical characteristics available in both datasets to evaluate the convergent and discriminant validity of the identified latent profiles.^{13,29} These included the Global Assessment of Functioning, rated from 0 to 100, with high scores indicating better functioning³⁰; the count of the number of diagnoses as a measure of complexity and burden; family history of bipolar disorder based on interview of the primary caregivers; and the Externalizing, Internalizing, and Attention Problems *T* scores on the Child Behavior Checklist (CBCL) using age and sex norms (mean = 50, SD = 10).³¹

Mood States at Intake. The Young Mania Rating Scale (YMRS)¹⁰ and Child Depression Rating Scale—Revised (CDRS-R)³² together characterized youth mood state at intake (Figure 1). CDRS-R scores less than 29 and YMRS scores less than 13 were labeled as euthymic; CDRS-R

FIGURE 1 Profile Plots for the 5 Groups on the 20 Mood Parcels, Sorted in Order of Decreasing Variation Between Profiles for the Hypomanic/Manic Parcels and Then the Depressive Symptom Parcels



Note: Profile labels: D = severe depression; md = hypomania and minor depression; Md = manic-level with moderate depressive symptoms; MD = manic-level with severe depression; mdsx = subthreshold symptoms of mania and depression.

greater than 29 and YMRS less than 13 were depressed; CDRS-R less than 29 and YMRS between 13 and 16.5 were hypomanic; CDRS-R greater than 29 and YMRS between 13 and 16.5 were mixed hypomanic; CDRS-R less than 29 but YMRS greater than 16.5 were manic; and CDRS-R greater than 29 and YMRS greater 16.5 were mixed.^{12,33}

Statistical Analyses

LPA using Mplus 8³⁴ identified homogenous subgroups based on similar response patterns to 20 continuous parcels from the PGBI. In all subsequent LPAs, indicator means were not assumed to be equal, whereas variances were held to be equal across profiles, and covariances among the indicators were fixed at zero across the profiles. We used the following model fit indices to compare models testing 1 to 9 profiles: Bayesian information criterion (BIC), Akaike information criterion (AIC), sample-size adjusted BIC (ABIC), the boot-strap likelihood ratio test (BLRT), the Lo–Mendell–Rubin likelihood ratio test (LMR), and entropy. Higher entropy suggests greater class separation.^{35,36} We prioritized the solution in which the BIC was lowest, entropy was highest, and the LMR was significant, indicating statistically significant improvement when comparing the k profiles and the model with $k+1$ profiles. We further considered whether profiles were substantively distinct and aligned with prior research as is recommended to avoid over-extracting latent profiles.^{35,37}

To quantify which parcels showed the greatest separation between profiles, we calculated both the eta-squared and the between-profile SD for each using profile membership as a factor. We looked at associations between profile membership and KSADS consensus diagnoses and other categorical variables using χ^2 analyses with Pearson standardized residuals as a way of detecting significant 1-df cells,³⁸ using Bonferroni correction. Dummy codes compared the means on continuous variables (such as mood severity interviews and global functioning) using the largest, lowest mood profile as the reference group. We also cross-classified mood profile membership by *a priori* conventional thresholds for depression, hypomania, mania, and mixed presentations on the YMRS and CDRS-R.^{12,33}

RESULTS

Demographic and Clinical Descriptive Statistics

A design goal was to increase generalizability and avoid selecting a model that was specific to one type of clinical setting. Preliminary analyses examined the degree to which the community and academic samples differed on

demographic and clinical characteristics. As expected, significant differences emerged between these 2 samples (Table 1; Figure S1, available online). The academic sample had a higher proportion of White youth (77% vs 6%) and higher rates of mood disorders (68% vs 41%) and bipolar disorder diagnoses (44% vs 12%), reflecting the mood disorders programs available at the center and the enriched referral streams, including partnership with an established adult mood disorders program. In contrast, the community sample presented with greater diagnostic comorbidity (mean = 2.63, SD = 1.35 diagnoses in community vs mean = 2.17, SD = 1.31 diagnoses per youth in the academic sample), as well as significantly higher rates of ADHD (65% vs 56%), ODD (37% vs 31%), CD (13% vs 8%), anxiety disorders (26% vs 14%), and PTSD (10% vs 3%).

These samples also diverged in severity on dimensional ratings. The academic sample scored higher on CBCL Internalizing, YMRS, and CDRS-R, indicating more severe mood-related symptoms. In contrast, the community sample scored higher on CBCL Externalizing and Attention problems, suggesting more pronounced behavioral difficulties. The heterogeneity observed in diagnoses and clinical characteristics across the datasets supported the potential for identifying generalizable latent profiles. Unlike prior work,¹³ we did not split the sample for the LPA because more severe manic and mixed mood profiles would be rarer in the community sample, so analyses of mood profiles would likely be underpowered.¹³ Pooling samples increased variance and representativeness, while also affording the opportunity to examine differences in prevalence across setting.

Latent Profile Analyses

When comparing models with 1 to 9 profiles, the BIC, AIC, and ABIC continually decreased until the addition of the ninth profile. Furthermore, the LMR and BLRT suggested significantly improved model fit when comparing the 3-profile solution to the 2-profile solution. Among models with 4 to 8 profiles, entropy was lowest in the 4-profile solution. The BLRT was significant for solutions modeling all profiles (Table S1, available online). Each of these fit indices has known limitations; thus, we visually inspected plots of the observed indicator means for distinct and meaningful profiles. Given these criteria, we selected the 5-profile solution as the best-fitting model (profile plots in Figure S3, available online).

Characterizing the Profiles

Using posterior probability estimates assigning youth to their most likely class, we compared the 5 identified profiles

TABLE 1 Demographics and Clinical Characteristics, Compared Across the Two Clinical Settings

	Combined (N = 1,369)	Academic (n = 758)	Community (n = 611)	Cohen d	p
Youth demographics					
Female	40%	40%	40%	0.00	NS
Age, y	11.06 (3.40)	11.41 (3.37)	10.63 (3.39)	0.23	****
White	625	77%	6%	1.91	****
Black	647	13%	88%	—	****
Hispanic	33	4%	1%	—	NS
Other	71	6%	4%	—	NS
Consensus diagnoses					
No. of Axis I KSADS dxs	2.38 (1.35)	2.17 (1.31)	2.63 (1.35)	−0.34	****
Any mood disorder dx	56%	68%	41%	0.56	****
Unipolar depressive dx	26%	24%	28%	−0.11	NS
Bipolar spectrum dx	30%	44%	12%	0.74	****
Any ADHD	60%	56%	65%	−0.19	***
Any substance use disorder	5%	6%	4%	0.07	NS
Any ODD	34%	31%	37%	−0.13	*
Any conduct disorder	10%	8%	13%	−0.16	*
Any anxiety disorder	19%	14%	26%	−0.31	****
Any PTSD	6%	3%	10%	−0.30	****
Other clinical characteristics					
GAF	53.78 (10.97)	54.59 (12.63)	52.82 (8.44)	0.16	**
CBCL Externalizing T	68.39 (11.00)	66.67 (11.69)	70.27 (9.86)	−0.33	****
CBCL Internalizing T	64.04 (11.00)	64.82 (11.46)	63.19 (10.42)	0.15	*
CBCL Attention T	68.48 (11.27)	67.06 (10.37)	70.03 (12.01)	−0.27	****
YMRS (raw sum)	9.34 (10.96)	11.89 (12.08)	6.05 (8.24)	0.55	****
CDRS-R (raw sum)	32.74 (15.13)	35.01 (16.26)	29.82 (12.98)	0.35	****
Family history of bipolar	34%	36%	33%	0.05	NS

Note: Data are mean (SD) or percentage. All effect sizes have been converted to Cohen d to provide a consistent metric. All are expressed as Academic sample minus Community sample (positive numbers mean that the Academic sample had the higher average or rate). A common convention is $d \sim 0.2$ is small, $d \sim 0.5$ is medium, and $d \sim 0.8$ is large. ADHD = attention-deficit/hyperactivity disorder; CBCL = Child Behavior Checklist; CDRS-R = Child Depression Rating Scale—Revised; dx = diagnosis; GAF = Global Assessment of Functioning; KSADS = Kiddie Schedule for Affective Disorders and Schizophrenia; NS = not significant; ODD = oppositional defiant disorder; PTSD = posttraumatic stress disorder; YMRS = Young Mania Rating Scale.

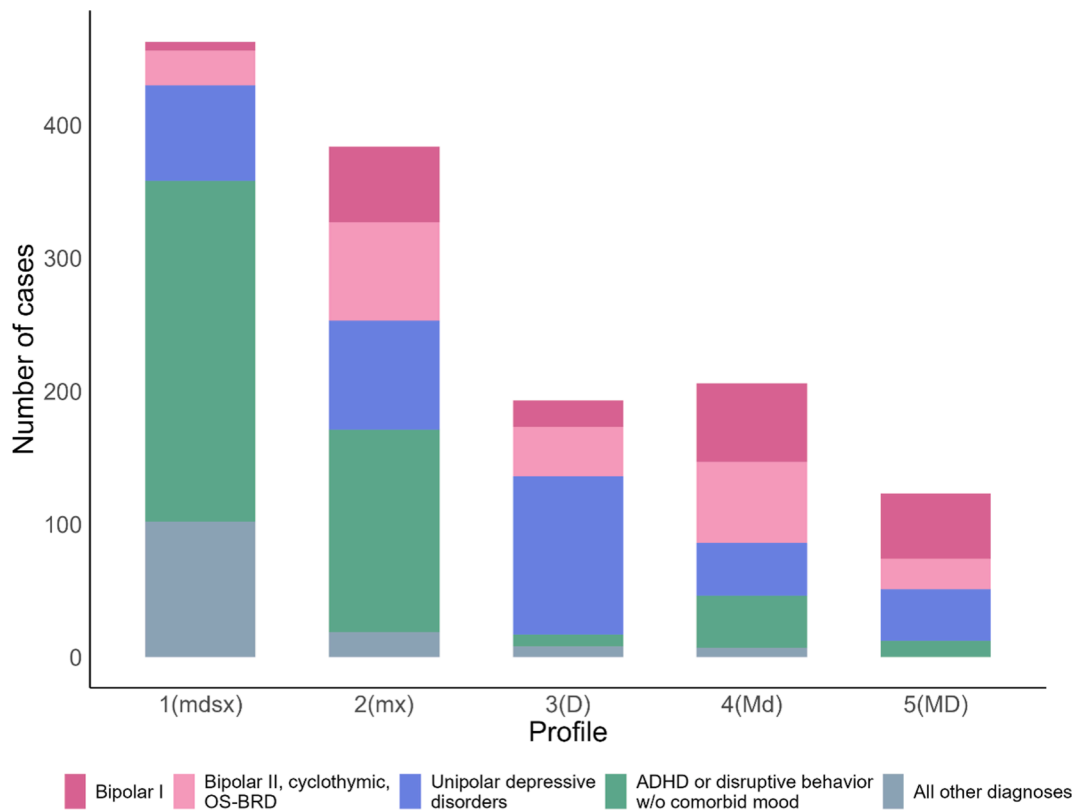
* $p < .05$, ** $p < .005$, *** $p < .0005$, **** $p < .00005$, two-tailed.

across parcels (Figure 2; Table S2, available online), diagnoses and demographics (Table 2; Figure S2, available online), diagnosis count and scale scores (Table 3), and mood state at intake (Figure 2; Table S3, available online). For categorical variables, we focus on when rates differed significantly from expectation for that profile (significant Pearson residuals). Dummy codes tested continuous variables differences using the largest profile group as the comparison.

Profile 1: Mild Overall Profile (mdsx) ($n = 463$). This was the most prevalent profile, characterized by the lowest scores across all PGBI parcels, indicating the mildest depressive and manic symptoms. This set had the lowest family risk for bipolar disorder ($p < .00005$),

lower-than-expected rates of diagnoses for depression ($p < .00005$) and bipolar ($p < .00005$) disorders, and the mildest mood symptoms (YMRS mean = 3.66 [0.46], CDRS-R mean = 24.24 [0.59], both $p < .00005$). Most members (71%) were categorized as euthymic at intake, although they met full criteria for at least one other non-mood psychiatric diagnosis.

Participants in this profile averaged 10.61 (± 0.15) years of age and had significantly fewer female and White participants ($ps < .00005$). They had lower-than-average rates of diagnoses of CD ($p < .05$) and ADHD ($p < .05$). This profile tended to have the fewest diagnoses (mean = 1.89 ± 0.06). They had significant Externalizing and Attention problems, and their average Internalizing problems, albeit lower, were still more than typical for their age and sex. Group members

FIGURE 2 Breakdown of Hierarchically Assigned Consensus Diagnoses of Mood Disorder Within the 5 Statistically Defined Latent Profiles (N = 1,369)

showed the highest global functioning (GAF mean = 59.45 $\pm$ 0.48, $p < .00005$), compared to all other profiles. Overall, this profile was the least impaired vs other profiles. Profile 1

served as the comparator for all subsequent profiles, both because it was the most common and because it showed low levels of mood symptoms despite consisting entirely of

TABLE 2 Comparison of Profiles on Rates of Diagnoses, Demographics, and Familial Risk of Bipolar Disorder, With Rates Shown as Deviations From Rate in Full Sample (N = 1,369)

Variable	n	Overall mean %	P1 (mdsx)	P2 (md)	P3 (D)	P4 (Md)	P5 (MD)	$\chi^2(4 df)$
Depression	352	26%	−10%****	−4%	+36%****	−6%	+6%	165.99****
Bipolar	413	30%	−23%****	+4%	−1%	+28%****	+28%****	243.66****
ODD	460	34%	−4%	+7%**	−15%****	+9%*	+3%	37.91****
CD	142	10%	−3%*	+1%	−5%	+10%****	+0%	32.29****
ADHD	820	60%	−5%*	13%****	−26%****	+15%****	−7%	106.13****
Substance use	67	5%	−1%	−2%	+4%*	−1%	+5%*	16.42**
Anxiety	259	19%	−3%	+0%	+4%	+0%	+5%	6.85
PTSD	79	6%	−1%	+0%	−2%	+1%	+6%*	9.76*
Familial risk	396	34%	−15%****	+6%*	−4%	+16%**	+22%**	61.28****
Female	547	40%	−3%	−7%**	+14%***	−2%*	+13%	33.93****
White	625	45%	−6%**	−6%*	+15%***	+7%	+10%	38.93****

Note: Significance for a specific profile is based on the standardized residual for that cell, per Agresti (2002).³⁸ P1 to P5 = profile 1 to profile 5, respectively. Profile labels: D = severe depression; md = hypomania and minor depression; Md = manic-level with moderate depressive symptoms; MD = manic-level with severe depression; mdsx = subthreshold symptoms of mania and depression. ADHD = attention-deficit/hyperactivity disorder; CD = conduct disorder; ODD = oppositional defiant disorder; PTSD = posttraumatic stress disorder.

* $p < .05$; ** $p < .005$; *** $p < .0005$; **** $p < .00005$, 2-tailed; Bonferroni corrected.

TABLE 3 Comparison of Profiles on Age, Diagnosis Count, Global Functioning, and Mood Symptom Severity (N = 1,369)

Variable	P1 (<i>mdsx</i>) (comparator)	P2 (<i>md</i>)	P3 (<i>D</i>)	P4 (<i>Md</i>)	P5 (<i>MD</i>)	F	R ²
Youth age, y	10.61 (0.15) ^a	−0.11 (0.23)	+2.22 (0.28)****	+0.17 (0.28)	+1.61 (0.33)****	23.05****	0.06
Diagnosis count	1.89 (0.06) ^a	+0.67 (0.09)****	+0.39 (0.11)***	+1.09 (0.11)****	+0.87 (0.13)****	32.92****	0.09
GAF	59.45 (0.48) ^a	−7.50 (0.71)****	−9.23 (0.90)****	−8.94 (0.86)****	−10.36 (1.04)****	55.37****	0.14
CBCL Externalizing T	62.62 (0.48) ^a	+8.10 (0.72)****	+4.35 (0.89)****	+12.28 (0.88)****	+12.06 (1.07)****	72.37****	0.19
CBCL Internalizing T	56.49 (43) ^a	+7.14 (0.65)****	+14.23 (0.81)****	+12.72 (0.79)****	+19.08 (0.96)****	160.29****	0.34
CBCL Attention T	63.30 (0.50) ^a	+6.87 (0.75)****	+3.90 (0.94)****	+11.41 (0.92)****	+11.51 (1.12)****	55.94****	0.15
YMRS	3.66 (0.46) ^a	+6.82 (0.69)****	+3.91 (0.85)****	+13.66 (0.83)****	+12.51 (1.00)****	88.18****	0.21
CDRS-R	24.24 (0.59) ^a	+6.20 (0.88)****	+22.94 (1.10)****	+10.49 (1.07)****	+21.69 (1.29)****	149.70****	0.31

Note: Profile 1 is used as the comparator (ie, reference group for four dummy codes), and the other groups are shown as deviations from the Profile 1 values. For example, Profile 1 has an average CDRS-R = 24.24, corresponding to very mild depression. Profile 5 has an average CDRS-R of 45.93 (=24.24+21.69), corresponding to moderate-to-severe depression. Profile labels: D = severe depression; md = hypomania and minor depression; Md = manic-level with moderate depressive symptoms; MD = manic-level with severe depression; *mdsx* = subthreshold symptoms of mania and depression. CBCL = Child Behavior Checklist; CDRS-R = Children's Depression Rating Scale—Revised, total score; GAF = Global Assessment of Functioning; YMRS = Young Mania Rating Scale total score.

^aAll values for Profile 1 are mean (SE) and for other groups are delta mean (SE).

* $p < .05$, ** $p < .005$, *** $p < .0005$, **** $p < .00005$, 2-tailed.

youths receiving outpatient mental and behavioral health services, making it a good “active control.” Because they often had mild, incidental mood symptoms, in the Angst proposed nomenclature, this would be the “*mdsx*” profile, only experiencing mood symptoms “within normal limits” on a psychiatric interview.³⁹ On a KSADS interview, these symptoms would be rated 1 or 2 (absent or subthreshold).²⁵

Profile 2: Moderate Overall Mood Profile (*md*) ($n = 384$). The second most common profile showed moderately elevated scores on all PGBI parcels. Youths in this profile reported slightly higher family risk for bipolar disorder (6% above the sample mean, $p < .05$), and their rates of depressive or bipolar disorder diagnoses were similar to expected rates based on the sample average. This group had hypomanic and mild-to-moderate depression symptoms, reflected by YMRS average 10.48 ± 0.69 , and CDRS-R average 28.84 ± 0.88 , both $p < .00005$ compared to Profile 1, mild/*mdsx*. Using *a priori* thresholds for mood severity, 36% of profile 2 members were euthymic, 10% “manic” (post hoc $p < .005$), 21% “mixed,” 9% hypomanic ($p < .00005$), and 25% depressed. There were significantly more manic and hypomanic cases in this profile than expected.

Demographics did not significantly differ from those in profile 1. Profile 2 members had significantly higher rates of ODD (7% above the mean, $p < .005$) and ADHD (13% above the mean, $p < .00005$), whereas rates of substance, anxiety, and PTSD diagnoses were not statistically different from the overall sample rates. Their average

number of diagnoses was 2.56 ($p < .00005$). They also had moderately lower global functioning, with a GAF decrement -7.50 ± 0.71 , $p < .00005$). Compared to the mild profile, it was associated with more comorbidity (~ 2.6 diagnoses per youth), Externalizing (mean = 70.7), Internalizing (mean = 63.6), and Attention problems (mean = 70.2), as well as worse global functioning. Profile 2 also had significantly more manic (YMRS mean = 10.5) and depressive (CDRS-R mean = 34.7) disorders. Of note, raters scored content on the mood interview only if it was episodic or represented a spontaneous exacerbation of other more chronic symptomatology, and did not “double count” symptoms across multiple KSADS modules without clinical evidence suggesting a “true” comorbidity.⁴⁰ This profile was the most heterogeneous in terms of diagnostic picture, including cases with bipolar or unipolar disorder but currently in more mild states (including some with spontaneous remission), plus cases with cyclothymic or other specified bipolar and related disorders. In the Angst proposed model, this profile would be labeled “*md*,” denoting a conceptual space that could contain “minor bipolar disorder” and “cyclothymic temperament” as well as cyclothymic disorder and prodromal (or partially remitting) major mood disorders.

Profile 3: Severe Depression Profile (*D*) ($n = 193$). Profile 3 was characterized by severe depression symptoms, with high scores on PGBI depression parcels but low-to-medium scores on manic item panels. This profile had the highest rate of depression (36% above the mean, $p < .00005$), yet only

average rate of bipolar disorder (1% below the mean, not significant). Mania symptom scores were slightly elevated ($+3.91 \pm 0.85$ YMRS, $p < .00005$), whereas depression symptom scores were the highest among all 5 profiles ($+22.94 \pm 1.10$ CDRS-R, $ps < .00005$). At intake, 63% of this group presented with a depressed mood based on present episode CDRS-R thresholds.

Members of this profile were significantly older ($+2.22 \pm 0.28$ years, $p < .00005$) and were more likely to be female ($p < .0005$) and White ($p < .00005$). They had a small but significantly higher rate of substance use disorder (4% above the mean, $p < .05$), while showing markedly lower rates of ODD (15% below the mean, $p < .00005$) and ADHD (26% below the mean, $p < .00005$). The average number of diagnoses was 2.28 ± 0.11 ($p < .0005$). This group exhibited significantly lower functioning than members of profile 1 (-7.50 ± 0.90 GAF, $p < .00005$).

Profile 4: Severe Mania Profile (Md) ($n = 206$). This profile had elevated PGBI mania parcels and moderate increases on PGBI depression panels. Youths in this set had a heightened family risk of bipolar (16% above the mean, $p < .005$) and significantly higher rates of bipolar diagnoses (28% above the mean, $p < .00005$). In contrast, the profile was not associated with unipolar depressive diagnoses (6% below the mean, not significant). This set showed the most severe manic symptoms ($+13.66 \pm .83$ YMRS, $p < .00005$) and moderately elevated depression ($+10.49 \pm 1.07$ CDRS-R, $p < .00005$). Based on *a priori* thresholds, 15% of cases presented with the manic mood state and 46% with a mixed mood state, consistent with the higher rates of bipolar disorder.

Profiles 4 and 1 did not differ significantly in age or other demographics. Profile 4 members were significantly more likely to have ODD (9% above the mean, $p < .05$), CD (10% above the mean, $p < .00005$), and ADHD (15% above the mean, $p < .00005$). The average number of diagnoses was the highest among all profiles at 2.98 ($p < .00005$). The global functioning was also significantly lower (-8.94 ± 0.11 GAF, $p < .00005$). Externalizing and Attention problems were significantly higher and averaged in the clinical range. Internalizing scores were significantly higher and averaged 69.2 (all $ps < .00005$).

Profile 5: Mixed Manic Depression Profile (MD) ($n = 123$). This was the smallest profile in terms of membership, characterized by severe elevations on both manic and depressive symptoms, with the highest scores on all PGBI parcels except increased energy. This set exhibited the greatest family risk of bipolar disorder (22% above the mean, $p < .005$) and was significantly more likely to be diagnosed

with bipolar disorders (28% above the mean, $p < .00005$). This profile showed the second highest level of manic and depressive symptoms ($+12.51 \pm 1.00$ YMRS, $+21.69 \pm 1.29$ CDRS, both $p < .00005$). Using *a priori* thresholds, 50% of the cases presented with a mixed mood state.

This group was, on average, older ($+1.61 \pm 0.33$ years, $p < .00005$), and it had a higher proportion of female participants ($p < .005$) and more White participants ($p < .005$). Youths in this profile had heightened risks of substance use (5% above the mean, $p < .05$) and PTSD (6% above the mean, $p < .05$). The average diagnosis count was 2.76 ± 0.13 ($p < .00005$). Profile 5 also displayed the worst global functioning (-10.36 ± 1.04 GAF, $p < .00005$). Externalizing, Internalizing, and Attention problems all averaged in the clinical range, with *T* scores in the mid-70s (all $ps < .00005$).

DISCUSSION

The goal of the present paper was to use latent profile analysis to group cases of youths seeking outpatient services for mental and behavioral health issues into sets with qualitatively different types of mood symptom presentation. Analyses used parent report of 20 different facets of mood symptoms, spanning from manic to mixed and depressive symptoms, and extending beyond the core set of *DSM* symptoms of mania or major depression.¹⁶ Analyses also pooled 2 different clinical referral streams, namely, an academic medical center and a community mental health clinic, to enhance the generalizability of findings. Results supported 5 profiles that showed strong associations with interview ratings of mood states and diagnoses, and several other markers of criterion validity, including differences in rates of family history of bipolar disorders. If these replicate, they provide an empirical approach to classification, potentially improving early detection of mood disorder types, and potentially offering a way of characterizing moderate and mixed mood presentations that have been challenging to fit into a *DSM/ICD* framework.

The largest profile (32% of the sample) had low levels of mood symptoms across the board. We provisionally call it the “mild mood” profile—mild to denote the symptom level (and also to avoid the potential connotation that “low mood” might imply significant depressive symptoms), and mild mood because all the profile members were still seeking outpatient services and carried at least 1 psychiatric diagnosis and other symptoms and impairment. In Angst²² parlance, the group had subthreshold hypomanic or depressive symptoms (*mdsx*), and using *a priori* thresholds,¹² most members would be considered euthymic or

within normal limits in terms of current mood severity. This group had significantly lower rates of consensus diagnoses of depression or bipolar disorders, lower interview ratings of current mood severity on the CDRS-R and YMRS, and less family history of mood disorder. The group was also characterized by significantly less conduct disorder or ADHD, which was more common in the CMHC sample—all of which could easily be attributed to differences in socioeconomic status and referral patterns for the CMHC vs academic medical center (Table 1).

The other 4 profiles all had moderate or severe levels of current parent-reported mood symptoms. However, they did not merely vary in regard to general level of symptoms: instead, profile plots crossed each other. The second-largest profile (29% of the sample) had moderate levels of mood symptoms on the PBGI, averaging between 0.2 and 1.2 points on a 0 to 3 response format per item parcel. Profile 2 was associated with significantly higher ADHD diagnoses compared to the full sample, with 73% of profile members affected. Profile 2 members were significantly more likely to be male, and the profile was more common in the community sample, but it was still associated with slightly more family history of mood disorder. Yet almost half of the members of this profile did not have a mood diagnosis and carried diagnoses only of ADHD and disruptive behavior, PTSD, or non-mood psychotic disorders.

Profile 3 had lower parent-reported scores on the hypomanic/manic parcels, higher scores on the depressive parcels, and correspondingly higher scores on the CDRS-R (averaging in the moderate-to-severe range), with low scores on the YMRS. The modal diagnosis was unipolar depression, and the modal mood state was “depressed” (*D*), although 15% of the profile members had sufficiently elevated YMRS scores to fall in the “mixed” range, and 15% were in the hypomanic plus depressed range (Angst “hypomixed”). From a methodological perspective, the presence of manic symptoms in this group is a function of the classification algorithm. Some of these could be “classification errors” due to measurement error in the PBGI scores, or sampling error. However, the fuzziness is also consistent with taxometric analyses generally finding no clear boundaries between qualitatively different groups in terms of depression or mania when these are analyzed as separate dimensions.⁴¹ The group still had notable Externalizing and Attention problems, although the Internalizing were the most elevated symptoms, despite having significantly lower rates of ADHD or ODD compared to the overall sample. Compared to the mild mood profile, there was significantly more comorbidity and lower global functioning; it was more common in female individuals and older youths.

The last 2 profiles were characterized by simultaneous elevations in both manic and depressive symptoms on the PBGI and the interview ratings. Most profile members met criteria for bipolar disorders, and both profiles tended to have the highest rates of comorbidity, the most severe mood presentations, and the worst overall functioning. The 2 groups differed in severity of depressive and internalizing symptoms, however. Profile 5 had CDRS-R scores almost identical to those of the unipolar depression profile (profile 3), whereas profile 4 had only moderate severity of depression. The 2 profiles had essentially equal YMRS scores. Using the Angst framework, profile 4 would be “*Md*” and profile 5 would be “*MD*.” Although parent-reported mood symptoms defined the groups, they showed more differences in terms of comorbid diagnoses, with the *Md* group having significantly higher rates of comorbid ADHD, ODD, and CD. The *MD* group was significantly older and had more substance use disorders. In the *DSM*, both would be labeled “mixed” or “mania with mixed features,” which would obscure not only the differences in severity of depression, but also the differences in clinical presentation and patterns of comorbidity.

Strengths of the present investigation include use of a large sample combining 2 distinct referral patterns, with significant differences in demographic and clinical features, enhancing the representativeness of the pooled sample and generalizability of findings. At the same time, both subsamples had the same measures available, and they shared interlocking rater training for the KSADS and mood severity ratings—an unusual feature that reduced methodological variance and increased internal validity. Another set of advantages flow from the focus on the PBGI, which has exceptionally broad content coverage that augments the core set of *DSM* symptoms of mania and depression, with 20 *a priori* symptom parcels that allowed a more fine-grained investigation of features differentiating profiles.

Limitations include that the analyses need to be replicated, ideally in a cohort followed longitudinally. The LPA fit indices did not all agree on a clear solution. Several of the statistical associations found in cross-sectional analyses will be important to disentangle with prospective designs, as many of the diagnoses follow different developmental trajectories (eg, externalizing symptoms decreasing vs mood and substance issues increasing with age). The sample size and diversity should make it somewhat more likely that an independent sample would find similar results. Because all enrollment sites were outpatient clinics, the severity of mood symptoms was also somewhat truncated. Individuals with floridly manic or acute mixed mood presentations would be more likely to be diverted to hospital or forensic settings. Profiles were based on a single

informant's perspective, and did not include clinical judgment. Parent-report has consistently shown better classification accuracy than teacher- or youth-report on mania scales,²⁸ but youth-reported depressive symptoms may show equal validity, particularly with older youths. The GBI has features that may limit feasibility in some settings, including a 12th grade reading level and complex items that embed symptoms in the context of changes in mood or energy.²⁸ These may increase burden, which has driven the development of short forms.²⁹ The 20 parcels require completion of the full 73 items, however.

Overall, present findings support several conclusions. Mild depression and varying severities of mixed mood presentation are clinically relevant—they are not rarities, and it may be possible to identify them clinically. Dimensional approaches accommodate this empirical reality. A corollary is that person-centered, cross-sectional profiling that simultaneously considers both depressive and manic symptoms might offer a framework for identifying individuals with different potential diagnoses, and qualitatively different patterns of comorbidity, warranting consideration of different interventions. Although clinicians are not going to use latent profile analysis to identify cases for targeted probing of diagnostic issues, software could be built to implement the classification algorithm and “out of sample” profile matching, much as is done in oncology and other areas of medicine. If profiles replicate, and particularly if they show treatment utility, then questionnaires completed at intake could facilitate more efficient diagnostic interviews and case formulation, realizing some of the potential for evidence-based assessment to catalyze improved outcomes.

CRedit authorship contribution statement

Eric A. Youngstrom: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yinuo Liu:** Writing – review & editing, Visualization, Formal analysis. **Amanda J. Thompson:** Writing – review & editing, Formal analysis, Conceptualization. **Kevin G. Stephenson:** Supervision, Formal analysis. **Jennifer Kogos Youngstrom:** Writing –

review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Robert L. Findling:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition.

This article is part of a special series of review and empirical articles devoted to examining dimensional alternatives to categorical diagnostic approaches for improving research insights and clinical practice in the field of child and adolescent neurodevelopmental and neuropsychiatric conditions. This series is edited by Guest Editors Mirko Uljarevic, MD, PhD, Robert Krueger, PhD, Eric Youngstrom, PhD, Andrew Whitehouse, PhD; Associate Editor Robert R. Althoff, MD, PhD; JAACAP Open Editor Manpreet K. Singh, MD, MS; and JAACAP Editor-in-Chief Douglas K. Novins, MD.

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ChatGPT 4o was used to annotate statistical code to make it more readable by other researchers, and NCH Enterprise ChatGPT was used to suggest ways of tightening phrasing to meet word count requirements.

Some results were previously presented at AACAP's 2024 Annual Meeting; October 14-19, 2024; Seattle, Washington.

Data Sharing: Sufficient data for replication of analyses available upon request from the authors.

Eric A. Youngstrom and Amanda J. Thompson served as the statistical experts for this work.

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